

1 Solution — GIAM 1.4

Problem 12

1.1 Problem

What is the value of $\lceil \pi \rceil^2 - \lceil \pi^2 \rceil$?

1.2 The Computation

Read the expression carefully: $\lceil \pi \rceil^2$ means $(\lceil \pi \rceil)^2$ — take the ceiling *first*, then square — while $\lceil \pi^2 \rceil$ squares *first*, then takes the ceiling. These are different operations in different orders.

First term, $\lceil \pi \rceil^2$. Since $\pi \approx 3.14159$,

$$\lceil \pi \rceil = \lceil 3.14159 \dots \rceil = 4, \quad \text{so} \quad \lceil \pi \rceil^2 = 4^2 = 16.$$

Second term, $\lceil \pi^2 \rceil$. Since $\pi^2 \approx 9.8696$,

$$\lceil \pi^2 \rceil = \lceil 9.8696 \dots \rceil = 10.$$

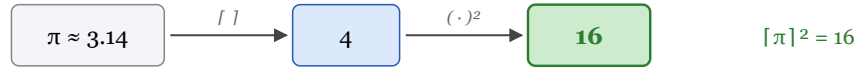
Subtract:

$\lceil \pi \rceil^2 - \lceil \pi^2 \rceil = 16 - 10 = 6.$
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1.3 Why It Isn't Zero — Order Matters

Ceiling and squaring do not commute — the order changes the result.

Route 1 — ceiling, then square:



Route 2 — square, then ceiling:



$$[\pi]^2 - [\pi^2] = 16 - 10 = 6$$

Figure 1.1: Route 1 (ceiling first): $\pi \rightarrow [\pi] = 4 \rightarrow 4^2 = 16$. Route 2 (square first): $\pi \rightarrow \pi^2 \approx 9.87 \rightarrow [\pi^2] = 10$. The two routes disagree, so $[\pi]^2 - [\pi^2] = 16 - 10 = 6$.

If ceiling and squaring *commuted*, the two terms would be equal and the answer would be 0. They do not. Rounding π up to 4 *before* squaring overshoots a lot — it jumps from 3.14 to 4, a gap of 0.86, which squaring then magnifies ($4^2 = 16$ versus $\pi^2 \approx 9.87$). Rounding *after* squaring only nudges 9.87 up to 10. The difference between “round-then-magnify” and “magnify-then-round” is the 6.

1.4 Remark — Ceiling Does Not Commute With Squaring

The lesson generalises: $\lceil x \rceil^2 \neq \lceil x^2 \rceil$ in general — the ceiling function does not pass through a squaring. For any $x \geq 0$ one *does* have the inequality

$$\lceil x \rceil^2 \geq \lceil x^2 \rceil,$$

because $\lceil x \rceil \geq x \geq 0$ forces $\lceil x \rceil^2 \geq x^2$, and $\lceil x \rceil^2$ is an *integer* that is $\geq x^2$, while $\lceil x^2 \rceil$ is the *least* such integer. The two are equal only in special cases (for instance when x is itself an integer); for $x = \pi$ the gap is a full 6. The same non-commuting behaviour afflicts square roots, logarithms, and most non-linear functions — rounding and a curved function simply don’t interchange.

1.5 Remark — The Floor Version Is Zero

A striking contrast: replace every ceiling with a floor and the answer collapses to **0**. Since $\lfloor \pi \rfloor = 3$ and $\pi^2 \approx 9.8696$,

$$\lfloor \pi \rfloor^2 - \lfloor \pi^2 \rfloor = 3^2 - \lfloor 9.8696 \rfloor = 9 - 9 = 0.$$

Here the two operations *happen* to agree — $\lfloor \pi \rfloor^2 = 9$ lands exactly on $\lfloor \pi^2 \rfloor = 9$ — purely because $3^2 = 9 \leq \pi^2 < 10$ keeps π^2 in the same integer interval $[9, 10)$ as $\lfloor \pi \rfloor^2$. This is a coincidence of where $\pi^2 \approx 9.87$ sits, not a general rule: $\lfloor x \rfloor^2 \leq \lfloor x^2 \rfloor$ always (for $x \geq 0$), but the gap need not be zero (e.g. $x = 3.9$: $\lfloor 3.9 \rfloor^2 = 9$ while $\lfloor 15.21 \rfloor = 15$, a gap of **6**). The floor problem giving **0** and the ceiling problem giving **6** together make the point vividly: *whether and by how much rounding commutes with squaring depends delicately on the number*.

1.6 Remark — Watch the Notation

The expression rewards careful reading of operator precedence, a recurring theme of these sections. The superscript binds to the *ceiling*, not to π :

$$\lceil \pi \rceil^2 = (\lceil \pi \rceil)^2 = 4^2 = 16, \quad \text{not} \quad \lceil \pi^2 \rceil = \lceil 9.87 \rceil = 10.$$

Misreading $\lceil \pi \rceil^2$ as $\lceil \pi^2 \rceil$ would make the whole expression $10 - 10 = 0$ — the wrong answer. The two bracket placements are *exactly* the two terms being subtracted, which is precisely what makes the problem a clean test of “do you see where the brackets go?” The intended value is **6**.